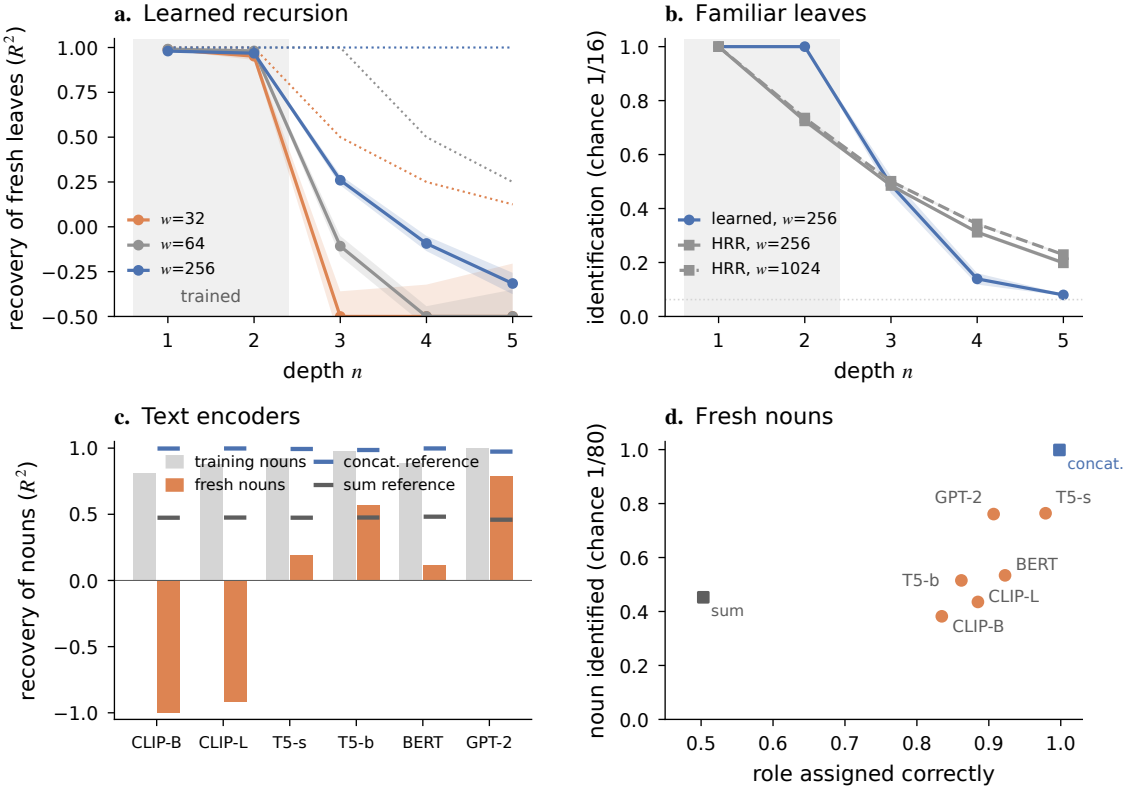

a. Learned recursion
recovery of fresh leaves (R^2)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
w=32
w=64
w=256
trained
1
2
3
4
5
depth n
b. Familiar leaves
identification (chance 1/16)
1.0
0.8
0.6
0.4
0.2
0.0
learned, w=256
HRR, w=256
HRR, w=1024
1
2
3
4
5
depth n
c. Text encoders
recovery of nouns (R^2)
1.0
0.5
0.0
−0.5
−1.0
training nouns
fresh nouns
concat. reference
sum reference
CLIP-B
CLIP-L
T5-s
T5-b
BERT
GPT-2
d. Fresh nouns
noun identified (chance 1/80)
1.0
0.8
0.6
0.4
0.2
0.0
concat.
GPT-2
T5-s
BERT
T5-b
CLIP-L
sum
CLIP-B
0.5
0.6
0.7
0.8
0.9
1.0
role assigned correctly